

Week 10-Documentation & Final Submission

Objective

The main objective of Week 10 is to complete the final version of the Word Scramble Game by combining all major game features into an attractive and interactive Java Swing GUI. The focus is on improving the game interface, adding difficulty levels, scoring, timer, hint system, and completing the overall game functionality.

Activities Completed

1. Developed the Final GUI Version

The final version of the Word Scramble Game was developed using Java Swing. The game was converted into a complete graphical interface with different panels, labels, buttons, text fields, and a difficulty selection system.

2. Added Difficulty Levels

Three difficulty levels were added to the game: **Easy, Medium, and Hard**. The player can select a difficulty level from the welcome screen before starting the game.

3. Added Separate Word Lists

Separate word arrays were created for each difficulty level. Easy contains simple words, Medium contains more complex words, and Hard contains advanced programming-related words.

4. Implemented Random Word Selection and Scrambling

The `Random` class was used to select a word randomly from the selected difficulty level. The `scramble()` method randomly rearranges the characters of the selected word before displaying it to the player.

5. Implemented Answer Checking

An answer validation system was implemented using the `checkAnswer()` method. The player's input is compared with the original word. A correct answer increases the score, while an incorrect answer displays a wrong-answer message.

6. Added Score and High-Score System

A scoring system was added based on difficulty level. Easy gives 10 points, Medium gives 20 points, and Hard gives 30 points for a correct answer. The highest score is stored and displayed as the high score.

7. Added 30-Second Timer

A 30-second countdown timer was implemented for each round using the Java Swing `Timer` class. When the time reaches zero, the timer stops and answer submission is disabled.

8. Implemented Hint System

A Hint button was added to help the player. The hint displays the first letter and total length of the correct word. Using a hint deducts 5 points when the player's score is sufficient.

9. Added 10-Round Game System

The game was designed with a total of 10 rounds. After each round, the player can move to the next word. When all 10 rounds are completed, the final result is displayed.

10. Added Game Control Buttons

Several control buttons were added to make the game easier to operate. These include **HINT**, **NEXT WORD**, **RESTART**, and **EXIT** buttons. Each button performs its specific game-related function.

11. Added Final Result Screen

A final result screen was implemented using `JOptionPane`. After completing the game, it displays the selected difficulty, final score, high score, and performance message.

12. Improved GUI Design

The GUI was enhanced using different colors, panels, borders, fonts, buttons, and labels. Separate colors were used for score, timer, difficulty, messages, and game controls to make the interface more attractive and user-friendly.

Deliverables

1. Complete Java source code (`week10.java`).
2. Final Word Scramble Game GUI.
3. Easy, Medium, and Hard word lists.
4. Working score and high-score system.
5. 30-second timer system.
6. Hint functionality.
7. 10-round game system.
8. Final result and performance display.
9. Restart, Next Word, and Exit functionality.
10. Final working version of the Word Scramble Game.

Expected Outcome

After completing Week 10, the final Word Scramble Game is expected to provide a complete and interactive gaming experience through a Java Swing GUI. The player will be able to select a difficulty level, solve scrambled words, earn points, use hints, play within a time limit, and complete 10 rounds. The final game should be visually attractive, user-friendly, and fully functional.